It can also be obtained by oxidising ammonia using bromine water

$$8NH_3 + 3Br_2 \longrightarrow 6NH_4Br + N_2$$

Properties

Nitrogen gas is rather inert. Terrestrial nitrogen contains 14.5% and 0.4% of nitrogen-14 and nitrogen-15 respectively. The later is used for isotopic labelling. The chemically inert character of nitrogen is largely due to high bonding energy of the molecules 225 cal mol^{-1} (946 kJ mol^{-1}). Interestingly the triply bonded species is notable for its less reactivity in comparison with other iso-electronic triply bonded systems such as -C≡C-, C≡O, X-C≡N, X-N≡C, -C≡C-, and -C≡N. These groups can act as donor where as dinitrogen cannot. However, it can form complexes with metal (M→ N≡N) like CO to a less extent

The only reaction of nitrogen at room temperature is with lithium forming Li_3N. With other elements, nitrogen combines only at elevated temperatures. Group 2 metals and Th forms ionic nitrides.

$$6Li + N_2 \longrightarrow 2Li_3N$$

$$3Ca + N_2 \xrightarrow{\text{red hot}} Ca_3N_2$$

$$2B + N_2 \xrightarrow{\text{bright red hot}} 2BN$$

Direct reaction with hydrogen gives ammonia. This reaction is favoured by high pressures and at optimum temperature in presence of iron catalyst. This reaction is the basis of Haber's process for the synthesis of ammonia.

$$\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons 2NH_3 \quad \Delta H_f = -46.2 \text{ kJ mol}^{-1}$$

With oxygen, nitrogen produces nitrous oxide at high temperatures. Even at 3473 K nitrous oxide yield is only 4.4%.

$$2N_2 + O_2 \longrightarrow 2N_2O \text{ (modify)}$$

Uses of nitrogen

1. Nitrogen is used for the manufacture of ammonia, nitric acid and calcium cyanamide etc.
2. Liquid nitrogen is used for producing low temperature required in cryosurgery, and so in biological preservation .

3.1.4 Ammonia (NH_3)

Preparation:

Ammonia is formed by the hydrolysis of urea.

$$NH_2CONH_2 + H_2O \longrightarrow 2NH_3 + CO_2$$

Ammonia is prepared in the laboratory by heating an ammonium salt with a base.

$$2NH_4^+ + OH^- \longrightarrow 2NH_3 + H_2O$$

$$2NH_4Cl + CaO \longrightarrow CaCl_2 + 2NH_3 + H_2O$$

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